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# PaperTrace: An Agentic Research Assistant for Academic Writing

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Group 52

## 1 Group members

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## 2 Project objective

Our objective is to develop **PaperTrace**, an end-to-end agentic research assistant that helps users explore research questions and develop evidence-grounded academic writing through dialogue and user's paper library. The agent will work on user-authorized workspace content and a user-selected paper library, while users retain control over literature selection, writing, and the acceptance of suggestions.

The system will support four connected objectives:

1. Search for relevant papers based on a research idea or draft, adjust search queries based on the research question, and help users build a selected paper library.
2. Support evidence-based question answering, writing discussions, and paper comparisons using the user-selected paper library.
3. Assist users in citing their writing by suggesting relevant references and supporting passages from the library.
4. After users have incorporated citations into their writing, check those citations against the cited papers for support, contradiction, missing qualifications, or overstatement, and suggest revisions when problems are identified.

PaperTrace will make task-dependent decisions about searching, retrieving evidence, and inspecting additional context. Evidence-based answers and citation judgments will include source locations or acknowledgements of insufficient evidence.

## 3 Motivation and challenges

## 4 Potential solutions

## 5 Experimental dataset description and evaluation plan